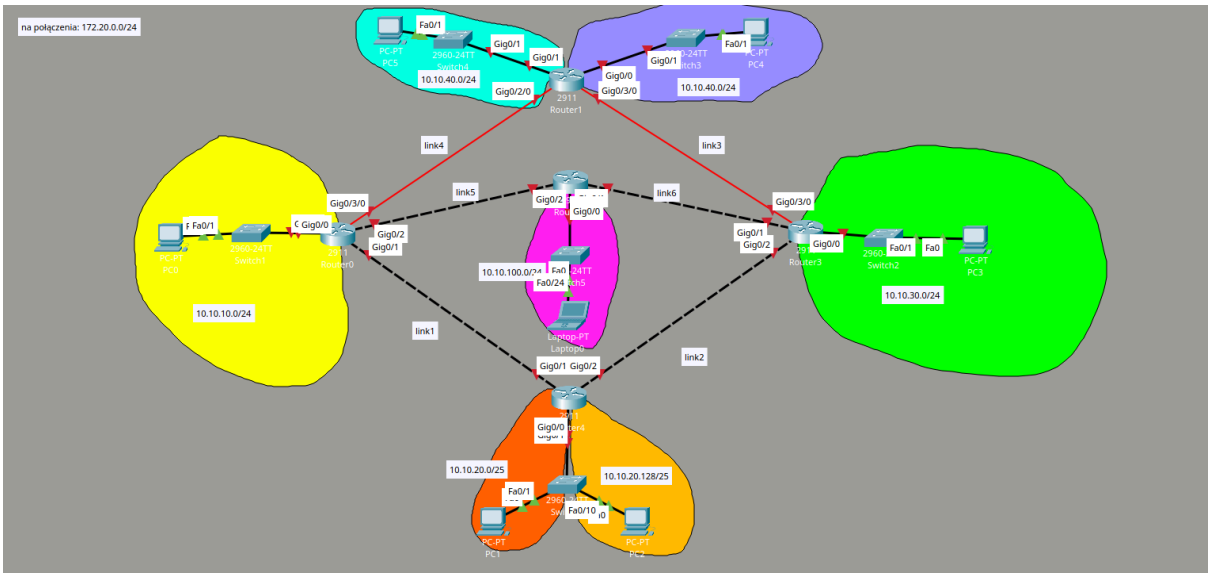
Cel:



Rozwiązanie:

1. Podzielenie sieci 172.20.0.0/24 na podsieci polaczeniowe

potrzebujemy 6 podsieci. robimy /30, bo PT ma problemy z p2p na /31

255.255.255.252

1. NET: .0 FH: .1 LH: .2 BR: .3
2. NET: .4 FH: .5 LH: .6 BR: .7
3. NET: .8 FH: .9 LH: .10 BR: .11
4. NET: .12 FH: .13 LH: .14 BR: .15
5. NET: .16 FH: .17 LH: .18 BR: .19
6. NET: .20 FH: .21 LH: .22 BR: .23

rozdajemy je według oznaczeń link (link1, link2, link3 itp)

2. Przypisanie adresów do interfejsów routerów

R0:

```
Router>en
Router#conft
Translating "conft"...domain server (255.255.255.255) % Name lookup aborted
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R0
R0(config)#no ip dom
R0(config)#no dom
R0(config)#no doman
R0(config)#no domai
R0(config)#exit
R0#
%SYS-5-CONFIG_I: Configured from console by console

R0#no doma
R0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#domain-look
R0(config)#domain-?
% Unrecognized command
R0(config)#ip domain?
domain  domain-lookup  domain-name
R0(config)#no ip domain-lookup
R0(config)#enable secret cisco
R0(config)#line console 0
R0(config-line)#exit
R0(config)#login
R0(config)#login admin secret ?
% Unrecognized command
R0(config)#login admin ?
% Unrecognized command
R0(config)#username admin secret ccna
R0(config)#line console 0
R0(config-line)#login local
R0(config-line)#exit
R0(config)#int g0/1
R0(config-if)#ip add 172.20.0.1 255.255.255.252
R0(config-if)#no shut
```

```
R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R0(config-if)#int g0/2
R0(config-if)#ip add 172.20.0.17 255.255.255.252
R0(config-if)#no shut

R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

R0(config-if)#int g0/3/0
R0(config-if)#ip add 172.20.0.13 255.255.255.252
R0(config-if)#no shut

%LINK-5-CHANGED: Interface GigabitEthernet0/3/0, changed state to down
```

```
R0(config)#service timestamps log datetime msec
```

R1:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable secret cisco
R1(config)#username admin secret ccna
R1(config)#line console 0
R1(config-line)#login local
R1(config-line)#exit
R1(config)#int g0/2/0
R1(config-if)#ip add 172.20.0.14 255.255.255.252
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2/0, changed state to up

R1(config-if)#exit
R1(config)#service tim
R1(config)#service timestamps log datetime
% Incomplete command.
R1(config)#service timestamps log datetime msec
R1(config)#int g0/3/0
R1(config-if)#ip add 172.20.0.9 255.255.255.252
R1(config-if)#no shut

%LINK-5-CHANGED: Interface GigabitEthernet0/3/0, changed state to down
R1(config-if)#
```

R2:

```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#enable secret cisco
R2(config)#username admin secret ccna
R2(config)#line console 0
R2(config-line)#login local
R2(config-line)#exit
R2(config)#service timestamps log datetime m
R2(config)#int g0/2
R2(config-if)#ip add 172.20.0.18 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
*Mar 01, 00:23:32.2323: %LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
*Mar 01, 00:23:32.2323: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up
R2(config-if)#int g0/1
R2(config-if)#ip add 172.20.0.21 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
*Mar 01, 00:24:21.2424: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

R3:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#enable secret cisco
R3(config)#username admin sec ccna
R3(config)#line con 0
R3(config-line)#login local
R3(config-line)#exit
R3(config)#int g0/3/0
R3(config-if)#ip add 172.20.0.10 255.255.255.252
R3(config-if)#no shut

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3/0, changed state to up

R3(config-if)#exit
R3(config)#service time log datetime m
R3(config)#int g0/1
R3(config-if)#ip add 172.20.0.10 255.255.255.252
% 172.20.0.8 overlaps with GigabitEthernet0/3/0
R3(config-if)#ip add 172.20.0.22 255.255.255.252
R3(config-if)#no shut

R3(config-if)#
*Mar 01, 00:19:51.1919: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
*Mar 01, 00:19:51.1919: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up
R3(config-if)#int g0/2
R3(config-if)#ip add 172.20.0.6 255.255.255.252
R3(config-if)#no shut

R3(config-if)#
*Mar 01, 00:20:07.2020: %LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
```

R4:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R4
R4(config)#service time log dat m
R4(config)#enable secret cisco
R4(config)#userna admin sec ccna
R4(config)#line con 0
R4(config-line)#login local
R4(config-line)#exit
R4(config)#int g0/1
R4(config-if)#ip add 172.20.0.2 255.255.255.252
R4(config-if)#no shut

R4(config-if)#
*Mar 01, 00:29:34.2929: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
*Mar 01, 00:29:34.2929: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up
R4(config-if)#int g0/2
R4(config-if)#ip add 172.20.0.5 255.255.255.252
R4(config-if)#no shut

R4(config-if)#
*Mar 01, 00:29:49.2929: %LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
*Mar 01, 00:29:49.2929: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up
```

3. teraz pora zajac sie poszczegolnymi subnetami. zaczniemy od tych z vlanami (router 1 i 4)

skoro juz jestesmy na R4 zacznijmy od jego subseci.

wchodzimy na jego switcha (S0):

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S0
S0(config)#enable secret cisco
S0(config)#us
S0(config)#us admin s ccna
S0(config)#li con 0
S0(config-line)#login local
S0(config-line)#exit
S0(config)#vlan 10
S0(config-vlan)#?
VLAN configuration commands:
  exit         Apply changes, bump revision number, and exit mode
  name         Ascii name of the VLAN
  no           Negate a command or set its defaults
  remote-span  Add the Remote Switched Port Analyzer (RSPAN) feature to the VLAN
S0(config-vlan)#name ## 10.10.20.0/25 ##
                     ^
% Invalid input detected at '^' marker.

S0(config-vlan)#
S0(config-vlan)#name ##10.10.20.0/25##
S0(config-vlan)#vlan 20
S0(config-vlan)#name ##10.10.20.128/25##
```

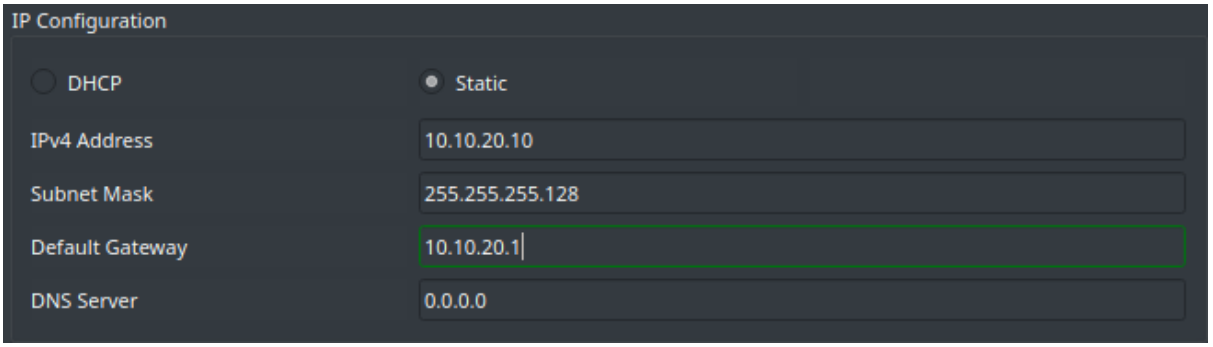
```
S0(config-vlan)#int f0/1
S0(config-if)#sw m acc
S0(config-if)#sw acc vlan 10
S0(config-if)#int f0/10
S0(config-if)#sw m acc
S0(config-if)#sw acc vlan 20
S0(config-if)#int g0/1
S0(config-if)#sw m trunk
S0(config-if)#sw trunk al
S0(config-if)#sw trunk allowed vlan
S0(config-if)#sw trunk allowed vlan 10,20
S0(config-if)#sw trnk nati
S0(config-if)#sw tr
S0(config-if)#sw trunk na
S0(config-if)#sw trunk native vl
S0(config-if)#sw trunk native vlan 88
```

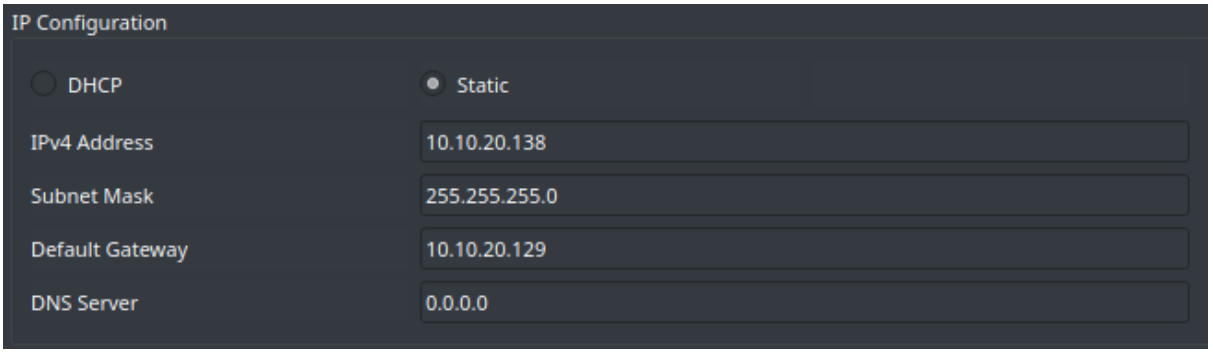
i teraz R4:

```
R4(config-if)#int g0/0.10
R4(config-subif)#encapsulation do
R4(config-subif)#encapsulation dot1Q
R4(config-subif)#encapsulation dot1Q ?
  <1-4094>  IEEE 802.1Q VLAN ID
R4(config-subif)#encapsulation dot1Q 10 ?
  native  Make this as native vlan
  <cr>
R4(config-subif)#encapsulation dot1Q 10
R4(config-subif)#ip add 10.10.20.1 255.255.255.128
R4(config-subif)#no shut
R4(config-subif)#int g0/0.20
R4(config-subif)#enc dot
R4(config-subif)#enc dot1Q 20
R4(config-subif)#ip add 10.10.20.129 255.255.255.128
R4(config-subif)#no shut
R4(config-subif)#int g0/0
R4(config-if)#no shut

R4(config-if)#
*Mar 01, 00:36:50.3636: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
*Mar 01, 00:36:50.3636: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
*Mar 01, 00:36:50.3636: %LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
*Mar 01, 00:36:50.3636: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10,
changed state to up
*Mar 01, 00:36:50.3636: %LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
*Mar 01, 00:36:50.3636: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20,
changed state to up
```

od razu konfiguracja pctow:





i test łączności z gatewayem i drugim pctem:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.20.1

Pinging 10.10.20.1 with 32 bytes of data:

Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.20.138

Pinging 10.10.20.138 with 32 bytes of data:

Request timed out.
Reply from 10.10.20.138: bytes=32 time<1ms TTL=127
Reply from 10.10.20.138: bytes=32 time=8ms TTL=127
Reply from 10.10.20.138: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.138:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>
```

klasa

4. teraz czas na konfiguracje reszty podsieci

R1:

```
User Access Verification

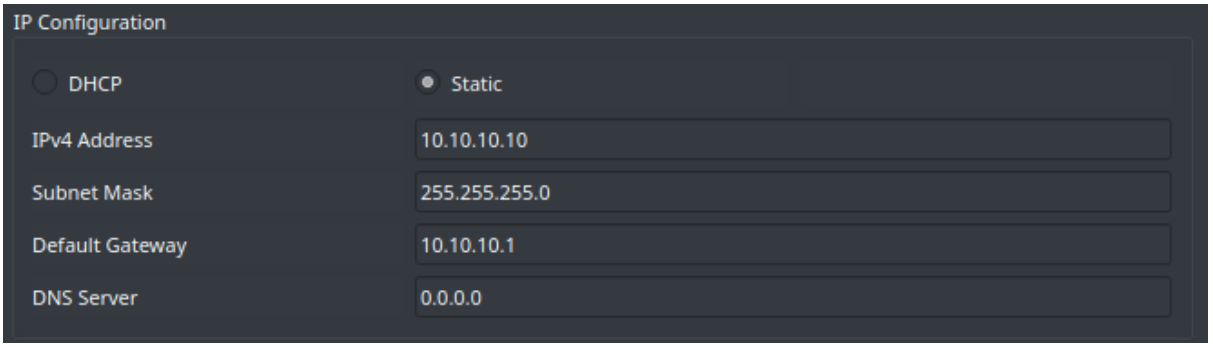
Username: admin
Password:

R0>en
Password:
R0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#ing g0/0
           ^
% Invalid input detected at '^' marker.

R0(config)#int g0/0
R0(config-if)#ip add 10.10.10.1 255.255.255.0
R0(config-if)#no shut

R0(config-if)#
*Mar 01, 00:33:58.3333: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
*Mar 01, 00:33:58.3333: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
```

PC0:

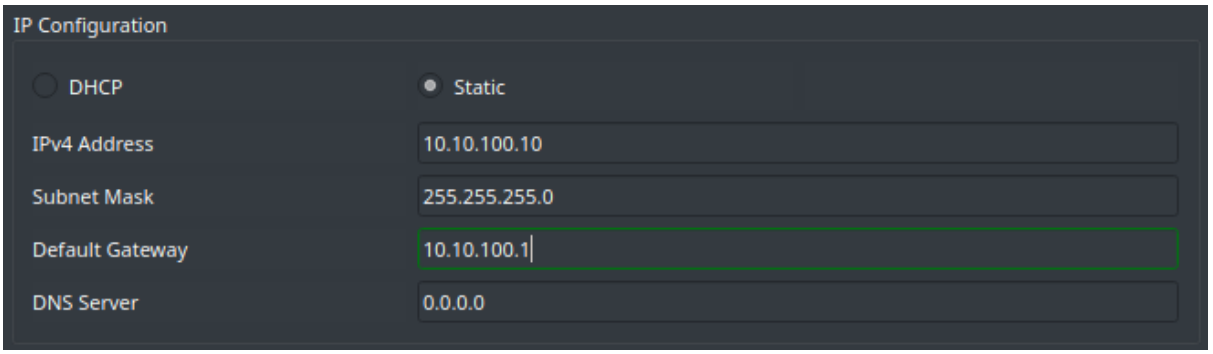


R2:

```
R2(config-line)#int g0/0
R2(config-if)#ip add 10.10.100.1 255.255.255.0
R2(config-if)#no shut

R2(config-if)#
*Mar 01, 00:43:41.4343: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
*Mar 01, 00:43:41.4343: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
```

Laptop0:

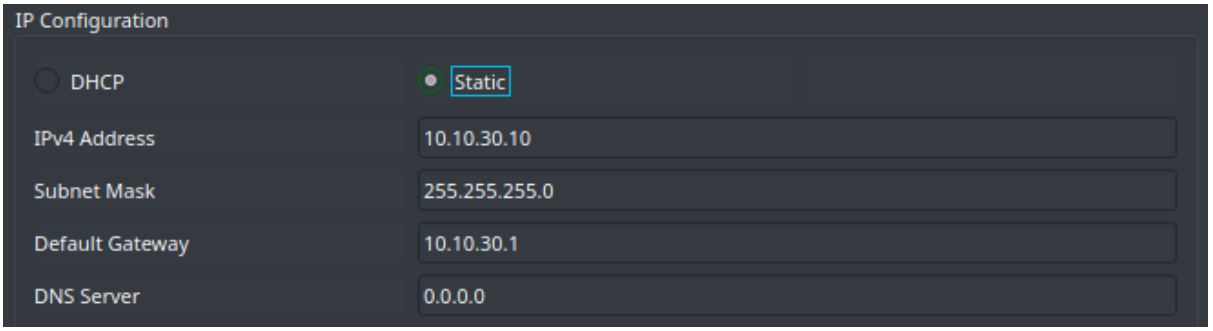


R3:

```
R3(config)#int g0/0
R3(config-if)#ip add 10.10.30.1 255.255.255.0
R3(config-if)#no shut

R3(config-if)#
*Mar 01, 00:36:54.3636: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
*Mar 01, 00:36:54.3636: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
```

PC3:



R1:

```
User Access Verification

Username: admin
Password:
% Login invalid

Username: admin
Password:

R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/0
R1(config-if)#ip add 10.10.40.1 255.255.255.0
R1(config-if)#no hsut
                ^
% Invalid input detected at '^' marker.

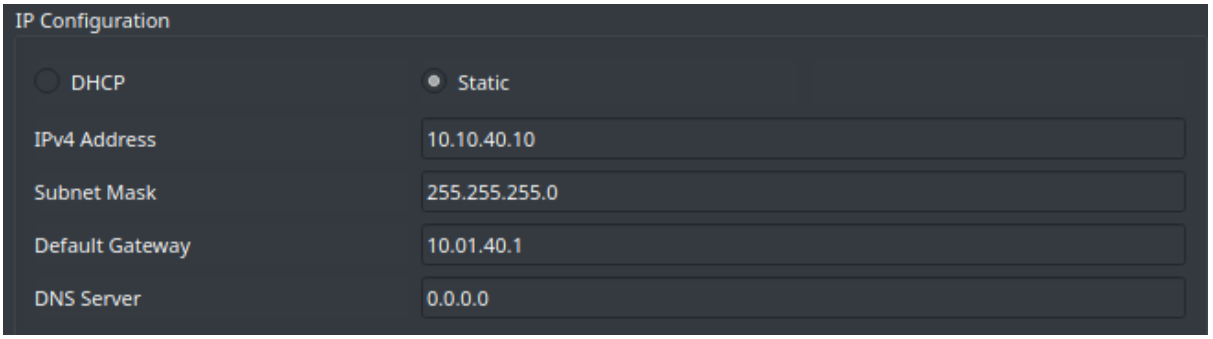
R1(config-if)#no shut

R1(config-if)#
*Mar 01, 00:38:21.3838: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
*Mar 01, 00:38:21.3838: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
R1(config-if)#int g0/1
R1(config-if)#ip add 10.10.40.1 255.255.255.0
% 10.10.40.0 overlaps with GigabitEthernet0/0
R1(config-if)#ip add 10.10.40.2 255.255.255.0
% 10.10.40.0 overlaps with GigabitEthernet0/0
R1(config-if)#ip add 10.10.50.2 255.255.255.0
R1(config-if)#no shut

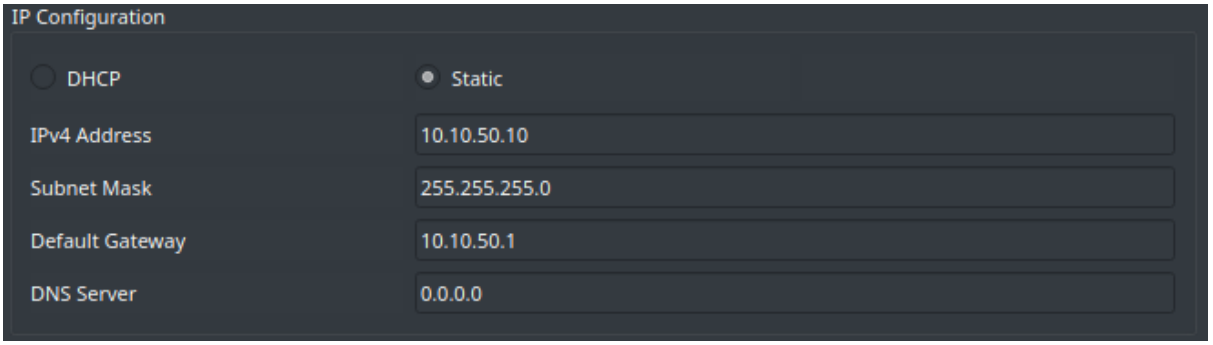
R1(config-if)#
*Mar 01, 00:39:43.3939: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
*Mar 01, 00:39:43.3939: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up
```

i tutaj najwidoczniej byl blad w zadaniu, zmienilem jedno 10.10.40.0 na 10.10.50.0

PC4:



PC5:



5. Ostatnia czesc konfiguracji - OSPF.

R0:

```
R0(config)#router ospf 1
R0(config-router)#passive-interface g0/0
R0(config-router)#network 10.10.10.0 0.0.0.255
% Incomplete command.
R0(config-router)#network 10.10.10.0 0.0.0.255 area 0
R0(config-router)#network 172.20.0.0 0.0.0.3 area 0
R0(config-router)#network 172.20.0.12 0.0.0.3 area 0
R0(config-router)#network 172.20.0.16 0.0.0.3 area 0
```

R1:

```
R1(config-if)#router ospf 1
R1(config-router)#passive-int g0/1
R1(config-router)#passive-int g0/0
R1(config-router)#network 10.10.50.0 0.0.0.255 area 0
R1(config-router)#network 10.10.40.0 0.0.0.255 area 0
R1(config-router)#network 172.20.0.12 0.0.0.3 area 0
R1(config-router)#network 172.20.0.8 0.0.0.3 area 0
```

R2:

```
R2(config-if)#router ospf 1
R2(config-router)#passive-int g0/0
R2(config-router)#network 10.10.100.0 0.0.0.255
% Incomplete command.
R2(config-router)#network 10.10.100.0 0.0.0.255 area 0
R2(config-router)#network 172.20.0.16 0.0.0.8 area 0
OSPF: Invalid address/mask combination (discontiguous mask)
R2(config-router)#network 172.20.0.16 0.0.0.3 area 0
R2(config-router)#network 172.20.0.20 0.0.0.3 area 0
R2(config-router)#
```

R3:

```
R3(config-if)#router ospf 1
R3(config-router)#passive-int g0/0
R3(config-router)#network 10.10.30.0 0.0.0.255 area 0
R3(config-router)#network 172.20.0.8 0.0.0.3 area 0
R3(config-router)#network 172.20.0.20 0.0.0.3 area 0
R3(config-router)#network 172.20.0.20 0.0.0.3 area 0
*Mar 01, 00:45:53.4545: 00:45:53: %OSPF-5-ADJCHG: Process 1, Nbr 172.20.0.14 on
GigabitEthernet0/3/0 network 172.20.0.20 0.0.0.3 area 0
*Mar 01, 00:46:03.4646: 00:46:03: %OSPF-5-ADJCHG: Process 1, Nbr 172.20.0.21 on
GigabitEthernet0/1 network 172.20.0.4 0.0.0.3 area 0
```

R4:

```
User Access Verification

Username:
Username: admin
Password:

R4>en
Password:
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router ospf 1
R4(config-router)#passive-int g0/0
R4(config-router)#network 10.10.20.0 0.0.0.127 area 0
R4(config-router)#network 10.10.20.128 0.0.0.127 area 0
R4(config-router)#network 172.20.0.0 0.0.0.3 area 0
R4(config-router)#network 172.20.0.4 0.0.0.3 area 0
R4(config-router)#
*Mar 01, 00:55:49.5555: 00:55:49: %OSPF-5-ADJCHG: Process 1, Nbr 172.20.0.17 on
GigabitEthernet0/1 from LOADING to FULL, Loading Done
*Mar 01, 00:55:54.5555: 00:55:54: %OSPF-5-ADJCHG: Process 1, Nbr 172.20.0.22 on
GigabitEthernet0/2 from LOADING to FULL, Loading Done
```

6. test łączności z PC0

```
C:\>ping 10.10.100.10

Pinging 10.10.100.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.100.10: bytes=32 time<1ms TTL=126
Reply from 10.10.100.10: bytes=32 time<1ms TTL=126
Reply from 10.10.100.10: bytes=32 time<1ms TTL=126

Ping statistics for 10.10.100.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Reply from 10.10.20.10: bytes=32 time<1ms TTL=126
Reply from 10.10.20.10: bytes=32 time<1ms TTL=126
Reply from 10.10.20.10: bytes=32 time<1ms TTL=126
Reply from 10.10.20.10: bytes=32 time<1ms TTL=126

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.20.138

Pinging 10.10.20.138 with 32 bytes of data:

Reply from 10.10.20.138: bytes=32 time<1ms TTL=126
Reply from 10.10.20.138: bytes=32 time<1ms TTL=126
Reply from 10.10.20.138: bytes=32 time<1ms TTL=126
Reply from 10.10.20.138: bytes=32 time<1ms TTL=126

Ping statistics for 10.10.20.138:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.30.10: bytes=32 time<1ms TTL=125
Reply from 10.10.30.10: bytes=32 time<1ms TTL=125
Reply from 10.10.30.10: bytes=32 time<1ms TTL=125

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

wiekszosc dziala, ale:

```
C:\>ping 10.10.50.10

Pinging 10.10.50.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.50.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.40.10

Pinging 10.10.40.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.40.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

wszystko na gorze nie dziala , wiec czas na

7. TROUBLESHOOTING

1. test ping do R1

```
C:\>ping 172.20.0.14

Pinging 172.20.0.14 with 32 bytes of data:

Reply from 172.20.0.14: bytes=32 time<1ms TTL=254
Reply from 172.20.0.14: bytes=32 time<1ms TTL=254
Reply from 172.20.0.14: bytes=32 time<1ms TTL=254
Reply from 172.20.0.14: bytes=32 time<1ms TTL=254

Ping statistics for 172.20.0.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

   idzie
2. test ping do wewnetrzego interfejsu R1

```
C:\>ping 10.10.50.1

Pinging 10.10.50.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.50.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

kaszka

3. sprawdzamy routing table R0

```
R0#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
C       10.10.10.0/24 is directly connected, GigabitEthernet0/0
L       10.10.10.1/32 is directly connected, GigabitEthernet0/0
O       10.10.20.0/25 [110/2] via 172.20.0.2, 00:13:11, GigabitEthernet0/1
O       10.10.20.128/25 [110/2] via 172.20.0.2, 00:13:11, GigabitEthernet0/1
O       10.10.30.0/24 [110/3] via 172.20.0.2, 00:13:01, GigabitEthernet0/1
                      [110/3] via 172.20.0.14, 00:13:01, GigabitEthernet0/3/0
                      [110/3] via 172.20.0.18, 00:13:01, GigabitEthernet0/2
O       10.10.40.0/24 [110/2] via 172.20.0.14, 00:15:55, GigabitEthernet0/3/0
O       10.10.50.0/24 [110/2] via 172.20.0.14, 00:15:55, GigabitEthernet0/3/0
O       10.10.100.0/24 [110/2] via 172.20.0.18, 00:16:54, GigabitEthernet0/2
     172.20.0.0/16 is variably subnetted, 9 subnets, 2 masks
C       172.20.0.0/30 is directly connected, GigabitEthernet0/1
L       172.20.0.1/32 is directly connected, GigabitEthernet0/1
O       172.20.0.4/30 [110/2] via 172.20.0.2, 00:13:01, GigabitEthernet0/1
O       172.20.0.8/30 [110/2] via 172.20.0.14, 00:15:03, GigabitEthernet0/3/0
C       172.20.0.12/30 is directly connected, GigabitEthernet0/3/0
L       172.20.0.13/32 is directly connected, GigabitEthernet0/3/0
C       172.20.0.16/30 is directly connected, GigabitEthernet0/2
L       172.20.0.17/32 is directly connected, GigabitEthernet0/2
O       172.20.0.20/30 [110/2] via 172.20.0.18, 00:16:54, GigabitEthernet0/2
```

mamy, czyli komunikat dobrze pokazuje: timed out a nie unreachable.

4. sprawdzenie interfejsow na R1

```
R1#show ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.40.1      YES manual up                    up
GigabitEthernet0/1     10.10.50.2      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
GigabitEthernet0/2/0   172.20.0.14     YES manual up                    up
GigabitEthernet0/3/0   172.20.0.9      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

okej, zle wpisane ip. pingowalem 10.10.50.1, a nie ma takiego. taki tez jest ustawiony jako DG w PC, wiec zmieniam to szybko

```
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/1
R1(config-if)#ip add 10.10.50.1 255.255.255.0
R1(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.40.1      YES manual up                    up
GigabitEthernet0/1     10.10.50.1      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
GigabitEthernet0/2/0   172.20.0.14     YES manual up                    up
GigabitEthernet0/3/0   172.20.0.9      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

5. ponowna proba pingow

```
C:\>ping 10.10.50.1

Pinging 10.10.50.1 with 32 bytes of data:

Reply from 10.10.50.1: bytes=32 time<1ms TTL=254
Reply from 10.10.50.1: bytes=32 time<1ms TTL=254
Reply from 10.10.50.1: bytes=32 time<1ms TTL=254
Reply from 10.10.50.1: bytes=32 time<1ms TTL=254

Ping statistics for 10.10.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.50.10

Pinging 10.10.50.10 with 32 bytes of data:

Reply from 10.10.50.10: bytes=32 time<1ms TTL=126
Reply from 10.10.50.10: bytes=32 time<1ms TTL=126
Reply from 10.10.50.10: bytes=32 time<1ms TTL=126
Reply from 10.10.50.10: bytes=32 time<1ms TTL=126

Ping statistics for 10.10.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.40.10

Pinging 10.10.40.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.40.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

okej, .50 sprawdzone, ale .40 nagle przestaje odpowiadac. pinguje

6. ping na .40.1

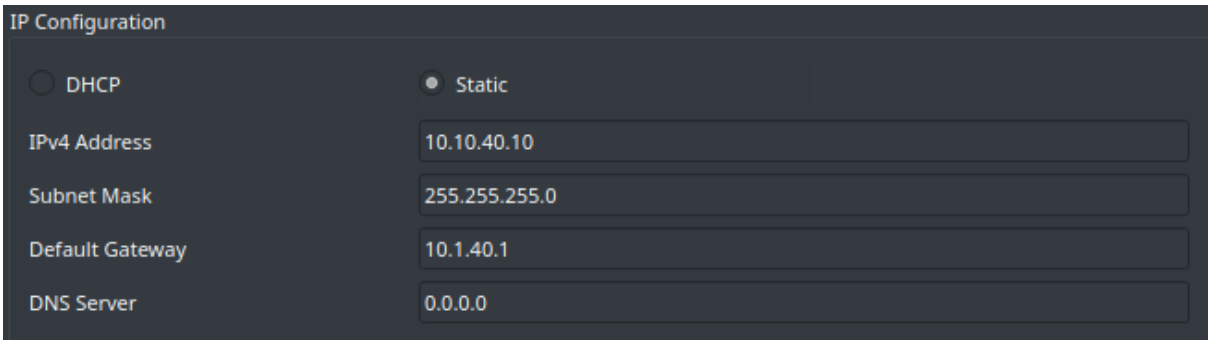
```
C:\>ping 10.10.40.1

Pinging 10.10.40.1 with 32 bytes of data:

Reply from 10.10.40.1: bytes=32 time<1ms TTL=254
Reply from 10.10.40.1: bytes=32 time<1ms TTL=254
Reply from 10.10.40.1: bytes=32 time<1ms TTL=254
Reply from 10.10.40.1: bytes=32 time<1ms TTL=254

Ping statistics for 10.10.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

idzie, sprawdzam w takim razie PC



i mamy nasz błąd! źle wpisany gateway

7. poprawiam i sprawdzam jeszcze raz pingi

```
C:\>ping 10.10.40.10

Pinging 10.10.40.10 with 32 bytes of data:

Reply from 10.10.40.10: bytes=32 time<1ms TTL=126
Reply from 10.10.40.10: bytes=32 time<1ms TTL=126
Reply from 10.10.40.10: bytes=32 time<1ms TTL=126
Reply from 10.10.40.10: bytes=32 time<1ms TTL=126

Ping statistics for 10.10.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

IDZIE!

wiec mamy wszystko, koniec laba :))

byl on dlugi i powtarzalny, ale dobrze jest miec juz ta pamiec miesniowa do podstawowych konfiguracji.